Assignment: SQL Tasks for SpecialForce Private Limited - Database Operations (MySQL)

Scenario:

SpecialForce Private Limited is expanding its workforce and needs help with managing its employee records, departments, and ongoing projects. As a fresh database consultant, your task is to create and manipulate the database to manage their growing employee, department, and project data.

Tasks:

Task 1: Create Tables

1. **Create three tables**: Employees, Departments, and Projects to track employees, departments, and projects, respectively.
    - Ensure each table has a **Primary Key** for uniquely identifying records.
    - Set up **Foreign Key** constraints to link employees to departments and projects.
    - Use appropriate **constraints** (e.g., NOT NULL, UNIQUE, etc.) to maintain data integrity.

Task 2: Insert Data (Given in excel sheet)

Once you have created the tables, insert the provided data into the respective tables. The data contains details about employees, departments, and projects.

Queries to Perform:

Query 1: Write a query to retrieve the first name, last name, and department name of all employees. If an employee does not belong to any department, the department name should be NULL.

Query 2: Write a query to find all employees in the IT department who earn more than ₹50,000.

Query 3: Write a query to list the first name, last name, and email of all employees whose first name starts with 'J' and whose email contains specialforce.com.

Query 4: Write a query to find all the distinct department names in the Departments table.

Query 5: Write a query to calculate the total salary expenditure of each department.

Query 6: Write a query to find the average salary of employees in the Finance department.

Query 7: Write a query to find the minimum and maximum salaries of employees in the Sales department.

Query 8: Write a query to count the number of employees in each department.

Query 9: Write a query to find all employees who were hired between January 1, 2018, and December 31, 2020. Sort the result by hire date in ascending order.

Query 10: Write a query to list all employees who do not have an email address.

Query 11: Write a query to find all employees who work in HR, Finance, or IT departments.

Query 12: Write a query to list the first name, last name, and salary of employees earning between ₹30,000 and ₹70,000. Sort the results by salary in descending order.

Transaction Management Tasks:

Use transaction control statements to manage the salary updates as follows:

Task 1: Increase HR Salaries:
Write a query to increase the salaries of all employees in the HR department by 5%. Start a transaction before applying the changes.

Task 2: Savepoint Before Sales Increase:
Set a savepoint before increasing the salaries of employees in the Sales department by 3%.

Task 3: Rollback Sales Salary Increase:
Rollback to the savepoint created before the Sales salary increase.

Task 4: Commit the Transaction:
After rolling back the Sales increase, commit the changes made to the HR department salaries.

Query 13: Write a query to join the Employees and Departments tables to list employees and their department names. Make sure all employees are included, even if they don't belong to any department.

Query 14: Write a query to list employees who are working on projects that started after January 1, 2023.

Query 15: Write a query to list all departments, even those without any employees assigned.

Query 16: Write a query to find the employee with the highest salary in each department.

Query 17: Write a query to remove all data from the Employees table but keep the structure intact.

Query 18: Write a query to drop the Projects table from the database.

Query 19: SpecialForce Private Limited realized they need to store the phone numbers of employees. Write a query to add a new column phone_number (VARCHAR(15)) to the Employees table using the ALTER statement.

Query 20: The company also decided to track the budget for each project. Write a query to add a column budget (DECIMAL(10,2)) to the Projects table.

Query 21: Write a query to find the 2nd largest salary from the Employees table using:

- A **subquery**.
- The LIMIT clause.

Query 22: Write a query to find the 3rd largest salary from the Employees table using:

- A **subquery**.
- The LIMIT clause.

Query 23: Write a query to drop the Projects table.

Query 24: Write a query to truncate the Employees table

Answers :-

Query 1:

```
mysql> Select Employee_Name , Department_Name from Employees e
   -> left join Departments d on
   -> e.Department_id = d.Department_id;
```

Query 2:

```
mysql> Select Employee_Name , Salary, Department_name from Employees e
   -> Inner join Departments d on
   -> e.department_id = d.department_id
```

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```
   -> where Department_Name = 'IT' and Salary > 50000;
```

Query 3:

```
mysql> Select Employee_Name , Email from Employees
   -> where Employee_Name like 'J%' and Email like '%specialforce.com%';
```

Query 4:

```
mysql> Select distinct Department_Name from Departments;
```

Query 5:

```
mysql> Select d.department_Name  , Sum(e.salary) as Total_Salary from
Departments d
   -> inner join Employees e on
   -> d.department_id = e.department_id
   -> group by d.department_id , d.department_name;
```

Query 6:

```
mysql> Select D.Department_Name , avg(e.salary) as Average_Salary from
Employees e
   -> Inner join Departments d on
   -> e.department_id = d.department_id
   -> where d.department_Name = 'Finance'
   -> group by D.department_Name ;
```

Query 7:

```
mysql> Select  d.department_name , min(e.Salary) as Minimum_Salary ,
max(e.salary) as Maximum_Salary from Employees e
   -> inner join Departments d on
   -> e.department_id = d.department_id
   -> where d.department_Name = 'Sales'
   -> group by d.department_Name;
```

Query 8:

```
mysql> Select d.department_Name , Count(e.employee_Name) as
Total_Employees from Departments d
   -> left join Employees e on
   -> d.department_id = e.department_id
   -> group by d.department_id , d.department_name;
```

Query 10:

```
mysql> Select Employee_Name , Email from Employees
   -> Where email = Null;
```

Query 11:

```
mysql> Select e.Employee_id , e.Employee_Name, d.department_Name from
Employees e
   -> inner join Departments d on
   -> e.department_id = d.department_id
   -> where d.department_name in ('HR' , 'Finance' , 'IT');
```

Query 12:

```
mysql> Select e.employee_name , e.salary from Employees e
   -> where salary between 30000 and 70000
   -> order by salary desc;
```

Task 1:

```
mysql> Update Employees  e
   -> inner join departments d on
   -> e.department_id = d.department_id
   -> set e.salary = e.salary * 1.05
   -> where d.department_name = 'HR';
Query OK, 3 rows affected (0.02 sec)
Rows matched: 3  Changed: 3  Warnings: 0
```

Task 2:

```
mysql> savepoint before_sales_increase;
```

Task 3:

```
mysql> Update Employees e
   -> inner join Departments d on
   -> e.department_id = d.department_id
   -> set e.salary = e.salary * 1.03
   -> where d.department_name = 'Sales';
Query OK, 2 rows affected (0.01 sec)
Rows matched: 2  Changed: 2  Warnings: 0
```

Task 4:

```
Rollback;
```

Task 5:

```
Commit;
```

Query 13:

```
mysql> Select  E.Employee_id ,E.employee_name , d.department_name from
Employees e
   -> left join departments d on
   -> e.department_id = d.department_id;
```

Query 15:

```
mysql> Select d.department_id , d.department_Name , e.Employee_id ,
e.Employee_name from Departments d
   -> left join Employees e
   -> on d.department_id = e.department_id;
```

Query 16:

```
mysql> Select e.Employee_name , e.salary , d.department_id ,
d.department_name from Employees e
   -> inner join departments d on
   -> e.department_id = d.department_id
   -> where e.salary = (
   -> select max(e2.salary) from Employees e2
   -> where e2.department_id = e.department_id
   -> );
```

Query 17:

TRUNCATE TABLE Employees; // Because is we Use Delete then the Primary Key Which is Auto Increment That Stops And if We use Truncate Then Only Data will be Deleted Whole Structure Remains Same and also That Auto_Increment is also Present .

Query 18:

```
Drop Projects;
```

Query 19:

```
mysql> alter table Employees
   -> add column phone_Number varchar(15);
```

Query 20:

```
mysql> alter table Projects
   -> add column budget decimal (10,2);
```

Query 21:

```
mysql> Select max(salary) as Second_higest_salary from Employees
   -> where Salary < (
   -> Select max(Salary) from Employees);
```

Query 22:

```
mysql> Select max(salary) as Third_higest_salary from Employees
   -> where Salary < (
```

```
   -> Select Max(salary) as Second_higest_salary from Employees
   -> where Salary < (
   -> Select max(salary) from Employees ));
```

Query 23:

```
mysql> drop Projects;
```

Query 24:

```
Mysql>  truncate Table  Employees;
```

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